(iii) $64x^2 - 81y^2 = \left(8^2 \times x^2\right) - \left(9^2 \times y^2\right) = (8x)^2 - (9y)^2$

$= (8x + 9y)(8x - 9y)$

Therefore, the factors of $64x^2 - 81y^2$ are $(8x + 9y)$ and $(8x - 9y)$.

Example 3.7 Factorise $9x^2 + 30xy + 25y^2$

Solution

$9x^2 + 30xy + 25y^2 = \left(3^2 \times x^2\right) + (2 \times 3 \times 5) \times xy + \left(5^2 \times y^2\right)$

$= (3x)^2 + 2 \times (3x) \times (5y) + (5y)^2$

This expression is in the form of identity $a^2 + 2ab + b^2 = (a+b)^2$.

Hence, $(3x)^2 + 2 \times (3x) \times (5y) + (5y)^2 = (3x + 5y)^2$

$= (3x+5y)(3x+5y)$.

Therefore, the factors of $9x^2 + 30xy + 25y^2$ are $(3x+5y)$ and $(3x+5y)$.

Example 3.8 Factorise $4x^2 - 4xy + y^2$

Solution

$4x^2 - 4xy + y^2 = 2^2 \times x^2 - (2 \times 2) \times xy + y^2$

$= (2x)^2 - 2 \times (2x) \times y + y^2$

This expression is in the form of identity $a^2 - 2ab + b^2 = (a-b)^2$.

Say $2x = a$ and $y=b$, thus we have,

$(2x)^2 - 2 \times (2x) \times y + y^2 = (2x - y)^2$

$= (2x - y)(2x - y)$

Therefore, the factors of $4x^2 - 4xy + y^2$ are $(2x - y)$ and $(2x - y)$.

Think

Can we factorise the following expressions using any basic identities? Justify your answer.

(i) $x^2 + 5x + 4$

(ii) $x^2 - 5x + 4$

Example 3.9 Factorise: $x^2 - 2xy + y^2 - z^2$

Solution

$x^2 - 2xy + y^2 - z^2 = (x^2 - 2xy + y^2) - z^2 = (x - y)^2 - z^2$ [by identity-3]

Let, $(x - y) = a$ and $z = b$.

Therefore, $(x - y)^2 - z^2 = a^2 - b^2$

We know that, $a^2 - b^2 = (a + b)(a - b)$ [Identity-4]

Hence, $x^2 - 2xy + y^2 - z^2 = (x - y + z)(x - y - z)$.

Therefore, the factors of $x^2 - 2xy + y^2 - z^2$ are $(x - y + z)$ and $(x - y - z)$.

7th_Maths_Algebra_Ch3_NEW.indd 60 05-11-2019 17:34:53